

ORF309/MAT380

Lifetimes II: Reliability

(pp. 84-89)

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Agenda for today

- More on serial and parallel connections
- Reliability
- Remaining lifetime
- Structure functions

PSET 5: due after break → focus on Midterm-1

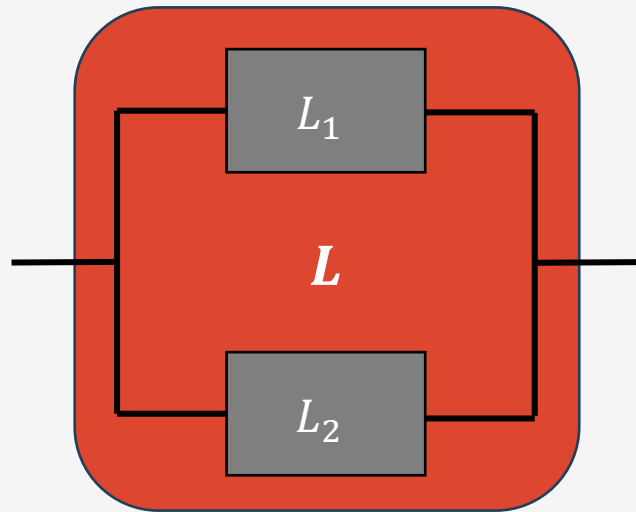
Rules & Info for Midterm-1 → see Yucheng's announcement!

**Midterm 1 practice tests:
now AVAILABLE on Canvas/Modules**

Midterm 1 will cover Chapters 1,2

Two components connected in parallel

$$L = \max\{L_1, L_2\} = L_1 \vee L_2$$



What is the Expected Lifetime of the device?

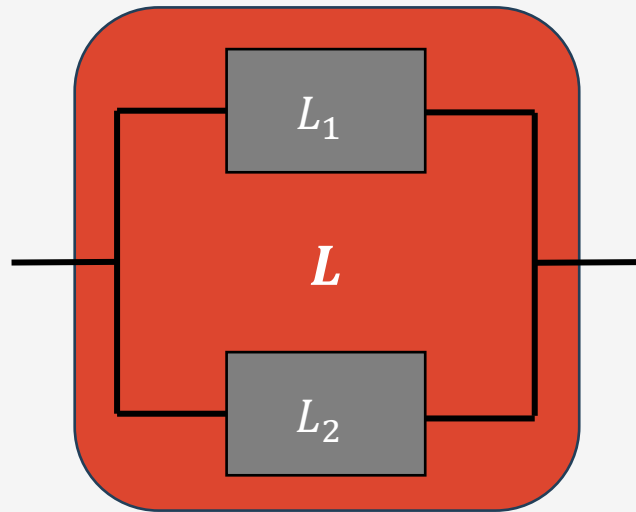
Note: $\forall x_1, x_2 \in R, x_1 \vee x_2 = x_1 + x_2 - x_1 \wedge x_2 \Leftrightarrow$

$\Leftrightarrow \forall x_1, x_2 \in R, \max\{x_1, x_2\} = x_1 + x_2 - \min\{x_1, x_2\}$

$$E[L] = E[L_1 \vee L_2] = E[\max\{L_1, L_2\}]$$

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$$E[L] = E[L_1 \vee L_2] = E[\max\{L_1, L_2\}]$$

$$E[L_1 + L_2 - L_1 \wedge L_2] =$$

$$E[L_1] + E[L_2] - E[L_1 \wedge L_2]$$

Special case: $L_1 \sim \text{Expon}(\mu)$ and $L_2 \sim \text{Expon}(\nu)$

$$E[L] = \frac{1}{\mu} + \frac{1}{\nu} - \frac{1}{\mu + \nu}$$

General way to compute $E[L]$ and *a useful trick*

- Let's develop a **more general** way to compute $E[L]$

$$E[L] = \int_0^{\infty} t P\{L \in dt\} = \int_0^{\infty} t \frac{d}{dt} P\{L \leq t\} dt = \int_0^{\infty} t \frac{d}{dt} (1 - P\{L > t\}) dt$$

- Computing the derivative within the integral may not be a happy experience...
- Some tricks can be really helpful:**

$$x = \int_0^x dt = \int_0^x 1 dt + \int_x^{\infty} 0 dt = \int_0^{\infty} \mathbf{1}_{\{x > t\}} dt$$

General way to compute $E[L]$ and a useful trick

- Replacing x with L we obtain:

$$L = \int_0^{\infty} \mathbf{1}_{\{L > t\}} dt$$

- And taking expectations on both sides:

$$E[L] = \int_0^{\infty} E[\mathbf{1}_{\{L > t\}}] dt = \int_0^{\infty} P\{L > t\} dt$$

No need to compute derivatives

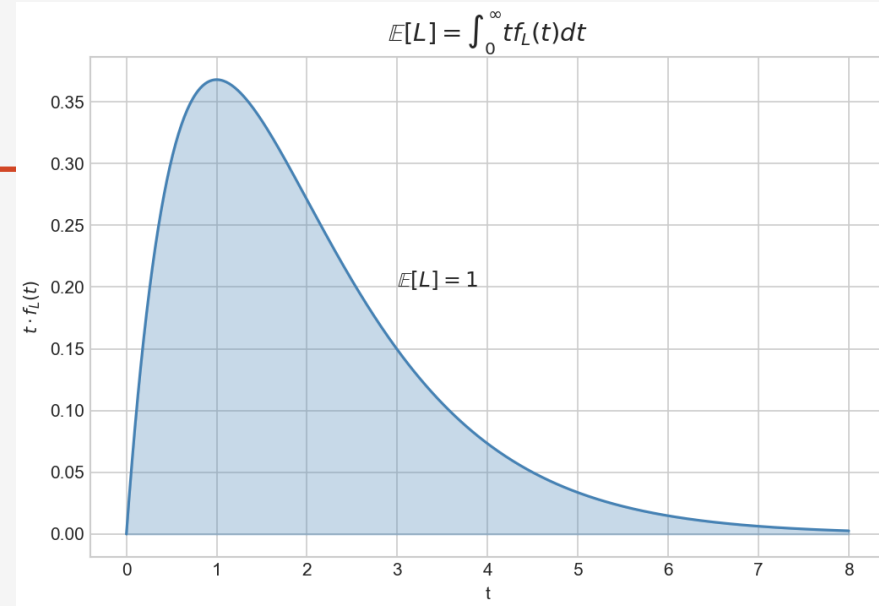
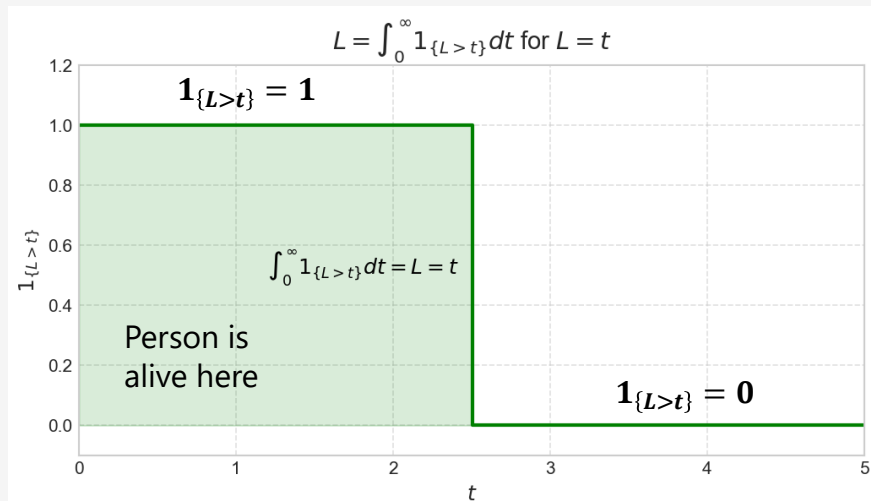
- **Remark:** This is a general trick and can be applied to any nonnegative RV

Visualizations

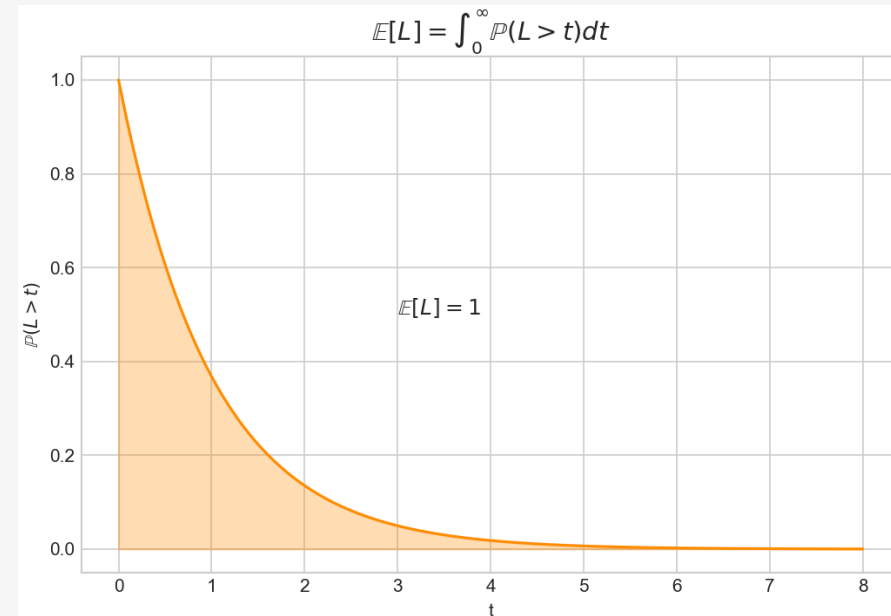
When $\mu = 1$.
For general case,
we would have

$$E[L] = \frac{1}{\mu}$$

One realization
when $L(\omega) = t$



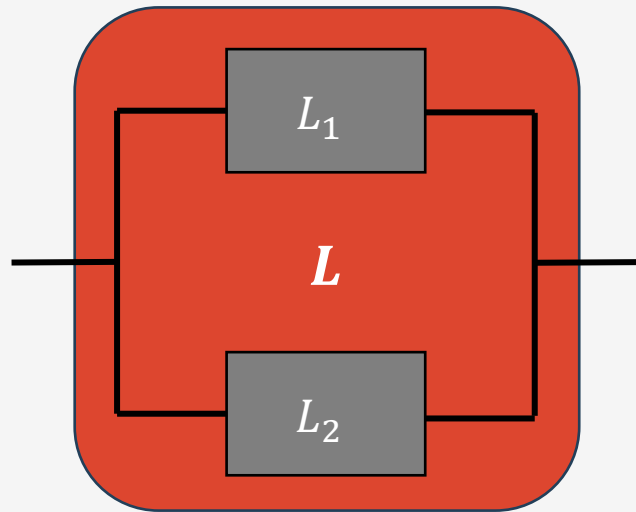
We measure $E[L]$ in
two different ways



Average of all
Realizations.
The area = 1

Two components connected in parallel – Expectation revisited...

$$L = \max\{L_1, L_2\} = L_1 \vee L_2$$



Let's apply the trick with expectations to our example

Assume: $L_1 \sim \text{Expon}(\mu)$ and $L_2 \sim \text{Expon}(\nu)$

Both memoryless components

$$\begin{aligned} \text{Then: } P\{L > t\} &= 1 - (1 - P\{L_1 > t\})(1 - P\{L_2 > t\}) \\ &= 1 - (1 - e^{-\mu t})(1 - e^{-\nu t}) = \\ &= e^{-\mu t} + e^{-\nu t} - e^{-(\mu+\nu)t} \end{aligned}$$

Using the previous trick: $E[L] = \int_0^\infty P\{L > t\} dt =$

$$= \int_0^\infty (e^{-\mu t} + e^{-\nu t} - e^{-(\mu+\nu)t}) dt = \frac{1}{\mu} + \frac{1}{\nu} - \frac{1}{\mu + \nu}$$

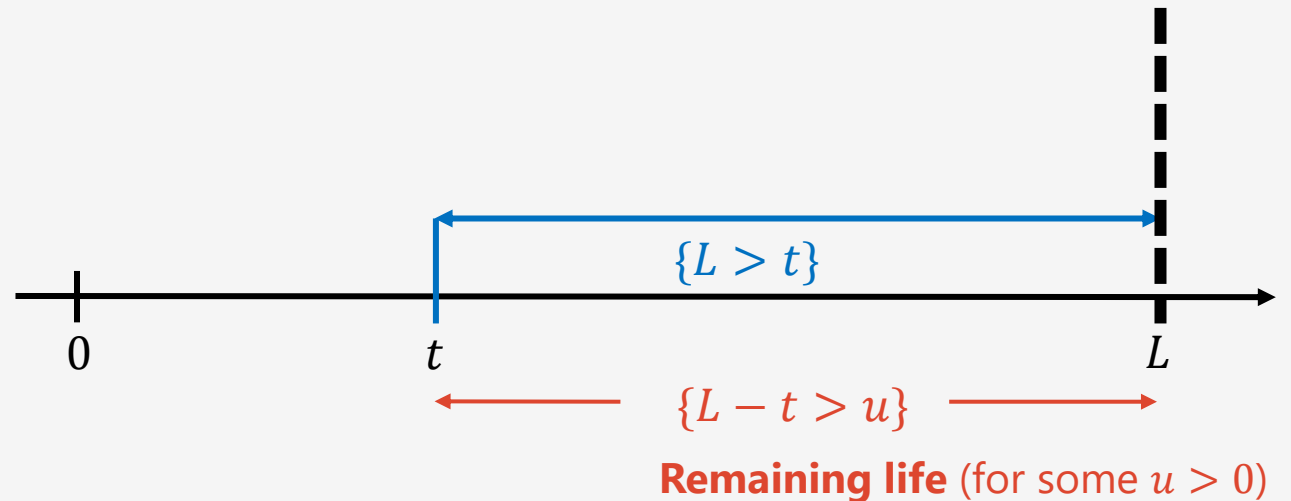
Remaining (useful) life

Expected remaining life of 2 components in parallel

- **Question:** How long do we expect to live, given that we made it to t years old?
- What **conditional expectation** are we looking for?

Expected remaining life of 2 components in parallel

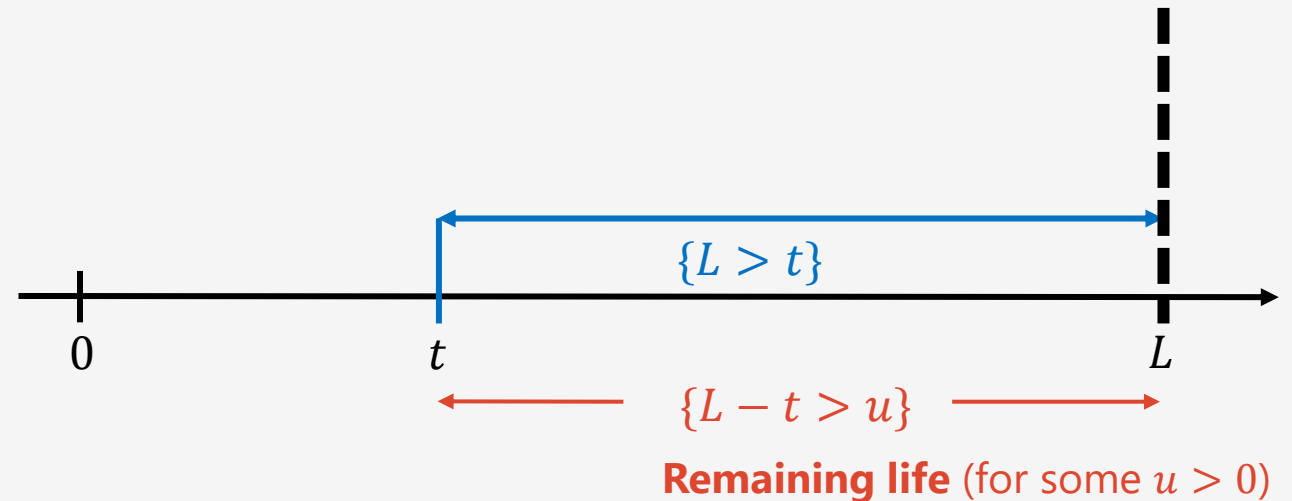
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Expected remaining life of 2 components in parallel

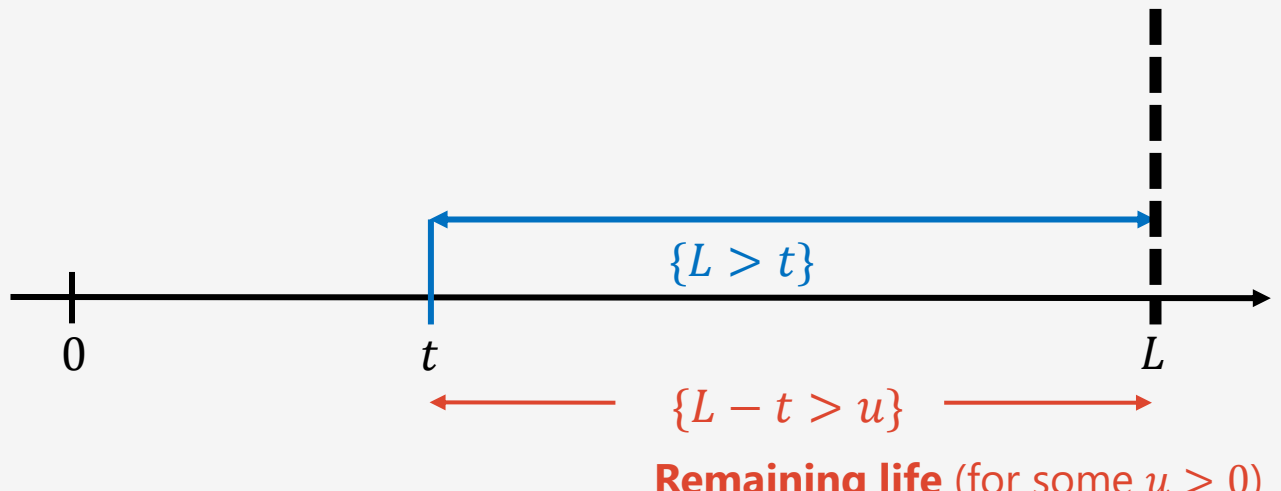
- **Question:** How long do we expect to live given that we made it to t years old?
- We need to compute the conditional expectation $E[L - t \mid L > t]$

$$E[L - t \mid L > t] = \int_0^{\infty} P\{L - t > u \mid L > t\} du = \int_0^{\infty} \frac{P\{L - t > u, L > t\}}{P\{L > t\}} du = \int_0^{\infty} \frac{P\{L - t > u\}}{P\{L > t\}} du =$$



Expected remaining life of 2 components in parallel

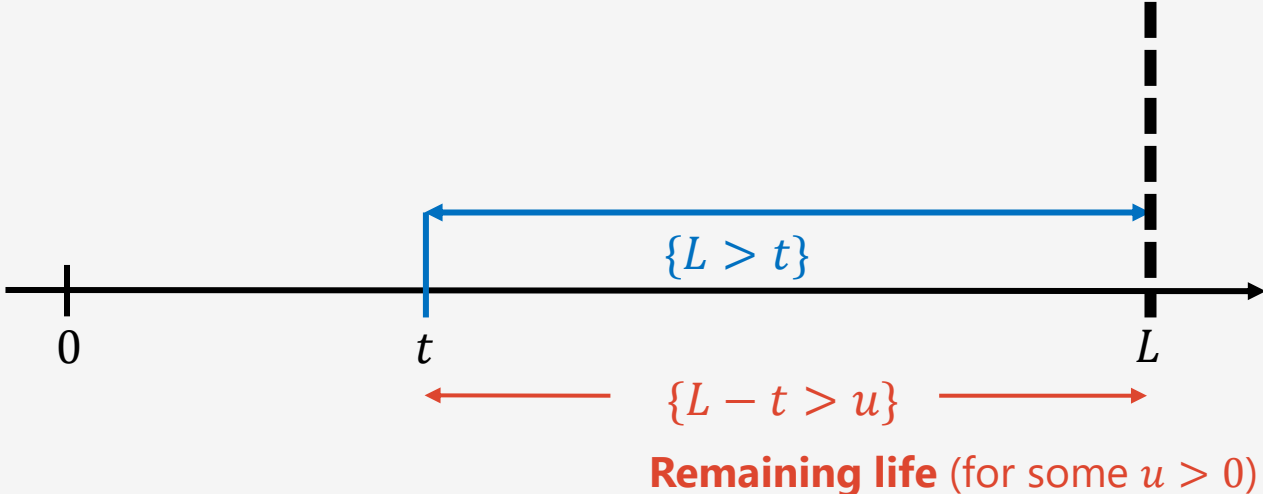
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Remaining life (for some $u > 0$)

Expected remaining life of 2 components in parallel

- **Question:** How long do we expect to live given that we made it to t years old?
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Expected remaining life of 2 components in parallel

$$\mu = \nu$$

- Consider the **special case**: $L_1 \sim \text{Expon}(\mu)$ and $L_2 \sim \text{Expon}(\mu)$ Both Memoryless

$$E[L - t | L > t] = \frac{\frac{2}{\mu}e^{-\mu t} - \frac{1}{2\mu}e^{-2\mu t}}{2e^{-\mu t} - e^{-2\mu t}} = \dots = \frac{1}{\mu} \left(1 + \frac{1}{4e^{\mu t} - 2} \right)$$

Observations: As t increases, the term in the parenthesis decreases.
Hence the expected remaining lifetime of the device decreases.

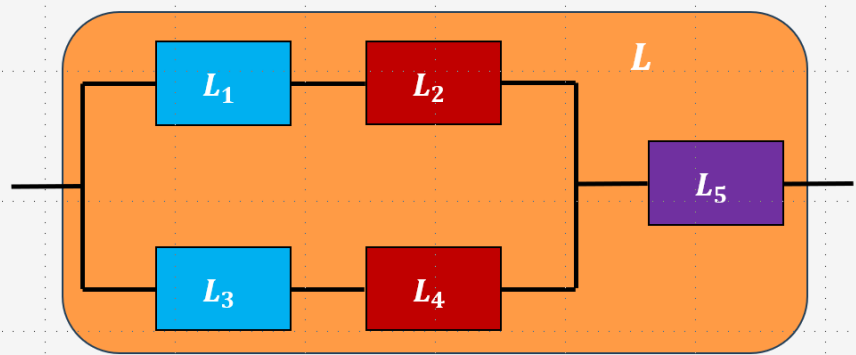
Although the two components are **memoryless**, the device that contains them is **aging**.

As time passes, **one of the components will die**, which affects the remaining lifetime of the device, since now it will only have one component.

Reliability

Reliability – structure functions

- **Structure function**: describes how a device is made up of its components



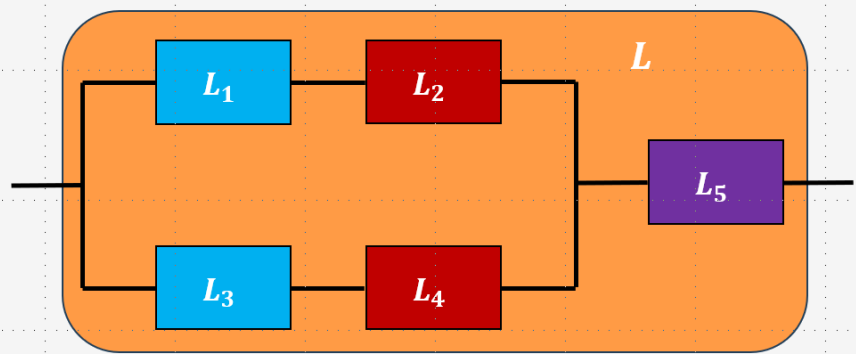
Lifetime of the system:

$$L = ((L_1 \wedge L_2) \vee (L_3 \wedge L_4)) \wedge L_5$$

This is not a very convenient way of doing computations.

Reliability – structure functions

- **Structure function:** describes how a device is made up of its components



Lifetime of the system:

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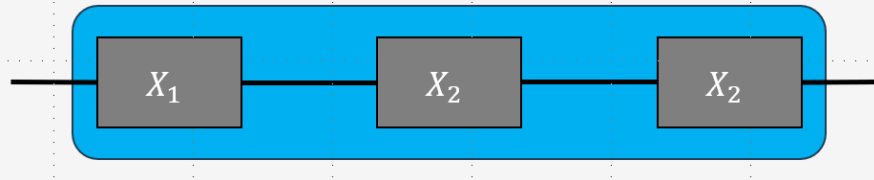
Define the **state variables**: $X_i = \begin{cases} 1, & \text{if component } i \text{ is } \textcolor{teal}{\text{alive}} \\ 0, & \text{is dead} \end{cases}$ for $i = 1, 2, 3, \dots$

$X_1, X_2, X_3, \dots$ are independent Bernoulli variables, with $P\{X_i = 1\} = p_i = E[X_i]$

Structure function: $\varphi: \{0,1\}^n \rightarrow \{0,1\}$ with $\varphi(X_1, X_2, \dots, X_n) = \begin{cases} 1, & \text{if the } \textcolor{brown}{\text{device}} \text{ is } \textcolor{teal}{\text{alive}} \\ 0, & \text{otherwise} \end{cases}$

Reliability – structure functions

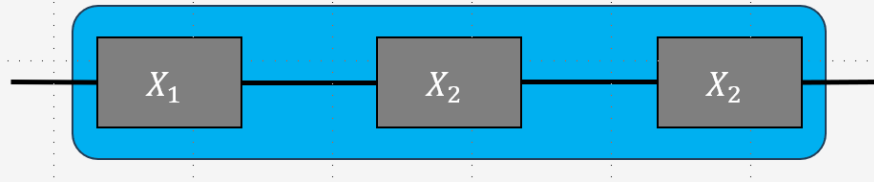
- **Example 1:** three components connected in series



Structure function: $\varphi(x_1, x_2, x_3) =$

Reliability – structure functions

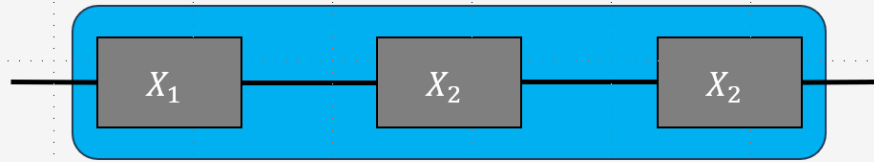
- **Example 1:** three components connected in series



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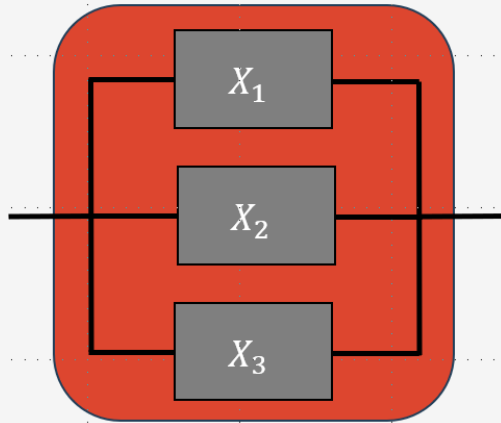
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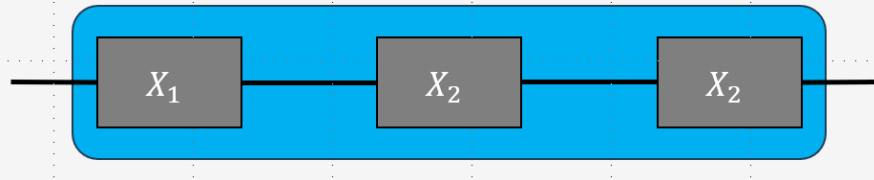
- **Example 2:** three components connected in parallel



Structure function: $\varphi(x_1, x_2, x_3) =$

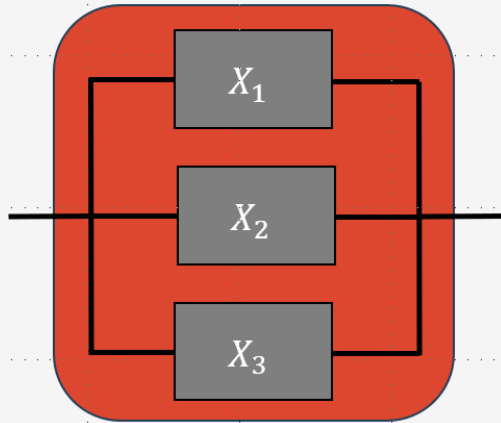
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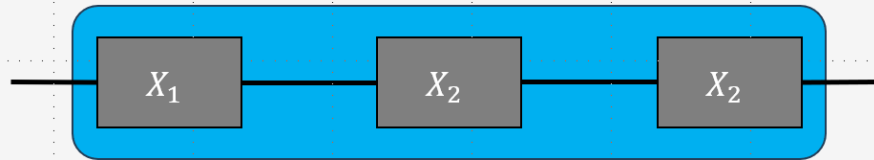
- **Example 2:** three components connected in parallel



Structure function: $\varphi(x_1, x_2, x_3) = 1 - (1 - x_1)(1 - x_2)(1 - x_3)$

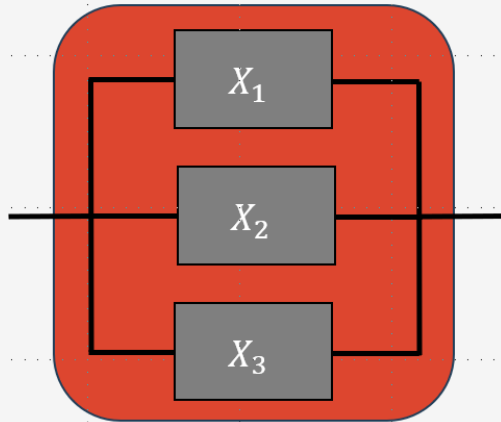
Reliability – structure functions

- **Example 1:** three components connected in series



Structure function: $\varphi(x_1, x_2, x_3) = x_1 x_2 x_3$

- **Example 2:** three components connected in parallel



Structure function: $\varphi(x_1, x_2, x_3) = 1 - (1 - x_1)(1 - x_2)(1 - x_3)$

NOTE: Assume you have **two** systems with n_1 and n_2 components with **structure functions** $\varphi_1(x_1, \dots, x_{n_1})$ and $\varphi_2(y_1, \dots, y_{n_2})$

If you connect them in **parallel** the structure function of the overall system is

$$\varphi(x_1, \dots, x_{n_1}, y_1, \dots, y_{n_2}) = 1 - \left(1 - \varphi_1(x_1, \dots, x_{n_1})\right) \left(1 - \varphi_2(y_1, \dots, y_{n_2})\right)$$

Structure functions – Probability & Expectation of being Alive

- We can use the **structure function** to express **probabilities & expectations**:

$$P\{\textit{system is alive}\} = P\{\overset{\text{\textit{\textcolor{blue}{\mathbf{\varphi is Bernoulli rv}}}}{\varphi}(X_1, \dots, X_n) = 1\} = E[\varphi(X_1, \dots, X_n)]$$

- **Note:** all the X_i are Bernoulli variables with $P\{X_i = 1\} = p_i$

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- Components in series:

$$P\{\textit{system is alive}\} = E[X_1, \dots, X_n] \stackrel{\text{series}}{=} p_1 \dots p_n = \varphi(p_1, \dots, p_n)$$

Structure functions – Probability & Expectation of being Alive

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- Components in parallel:

$$\begin{aligned} P\{\textit{system is alive}\} &= E[\varphi(X_1, \dots, X_n)] = \\ &= E[1 - (1 - X_1) \dots (1 - X_n)] \stackrel{\text{parallel}}{=} 1 - (1 - p_1) \dots (1 - p_n) = \varphi(p_1, \dots, p_n) \end{aligned}$$

Structure functions and Probability & Expectation of being Alive

- Hence the formula **links** structure functions with Lifetimes and Probability/Expectation of the Bernoulli variables

$$P\{\text{system is alive}\} = E[\varphi(X_1, \dots, X_n)] = \varphi(p_1, \dots, p_n)$$

- ! This only holds for systems of independent components that are **connected in series or in parallel** !
- CAUTION:** in general, for any function φ and random variables $X_1, \dots, X_n$ we have:

$$E[\varphi(X_1, \dots, X_n)] \quad ?? \quad \varphi(p_1, \dots, p_n)$$

Structure functions and Probability & Expectation of being Alive

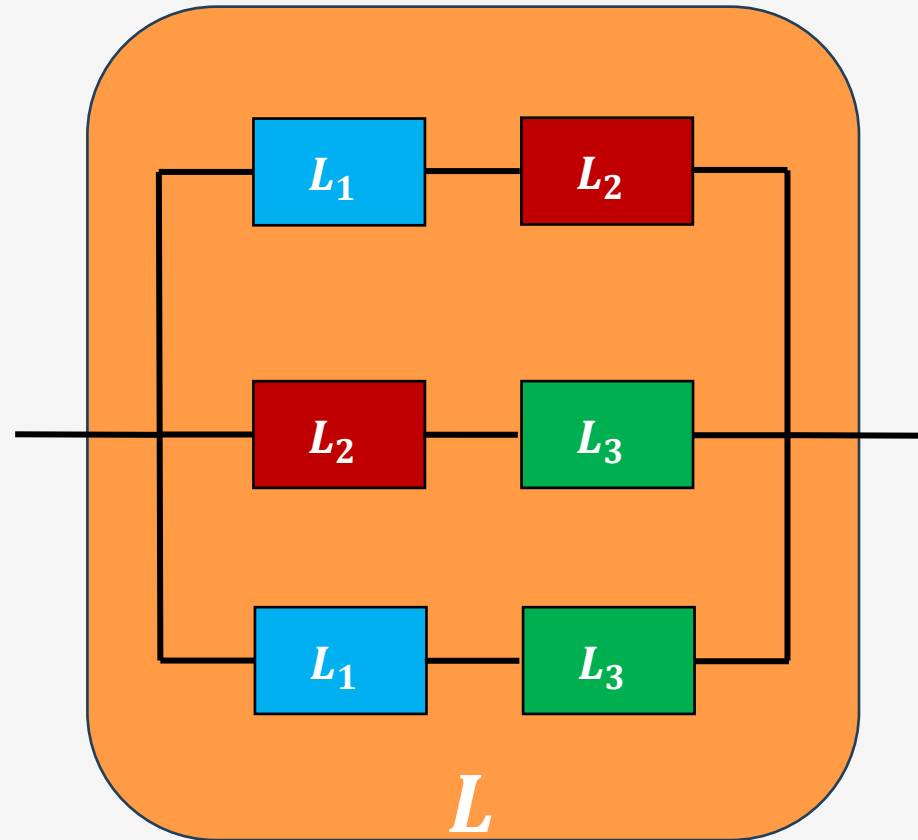
- **Counter-Example:** (for $E[\varphi(X_1, \dots, X_n)] \stackrel{??}{=} \varphi(p_1, \dots, p_n)$)
 - A **system of three components** is **alive** if **2-out-of-3 components** are **alive**
 - This system **cannot** be described by series or parallel connections
 - Structure function: $\varphi(x_1, x_2, x_3) = 1 - (1 - x_1x_2)(1 - x_2x_3)(1 - x_1x_3)$
 - Note: $1 - X_1X_2 \not\sqsubseteq 1 - X_2X_3 \not\sqsubseteq 1 - X_1X_3$

$$E[\varphi(X_1, X_2, X_3)] = E[1 - (1 - X_1X_2)(1 - X_2X_3)(1 - X_1X_3)] = \dots$$

$$\dots = E[X_1X_2 + X_2X_3 + X_1X_3 - 2X_1X_2X_3] = p_1p_2 + p_2p_3 + p_1p_3 - 2p_1p_2p_3 \neq \varphi(x_1, x_2, x_3)$$

Expand φ to see it...

2 out of 3 system diagram



Lifetime of a system of components

- So far, we have explored: $P\{\text{system is alive}\}$
- **How about the Lifetime of the *system* in terms of its *components*??**
- Define the **state variables** of the *system* and its *components*

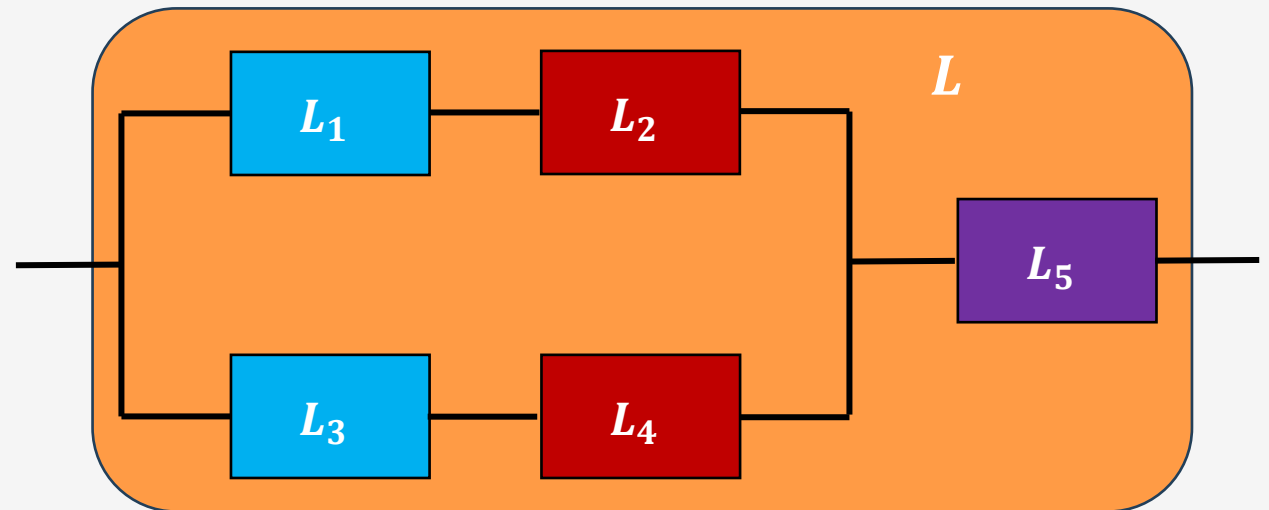
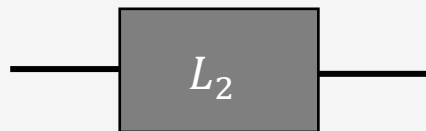
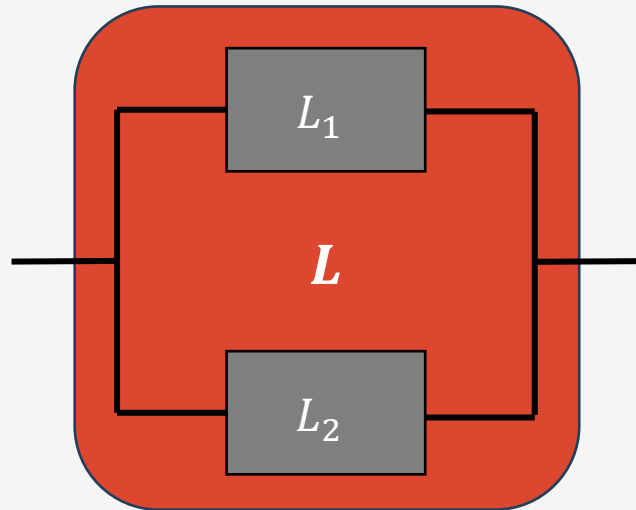
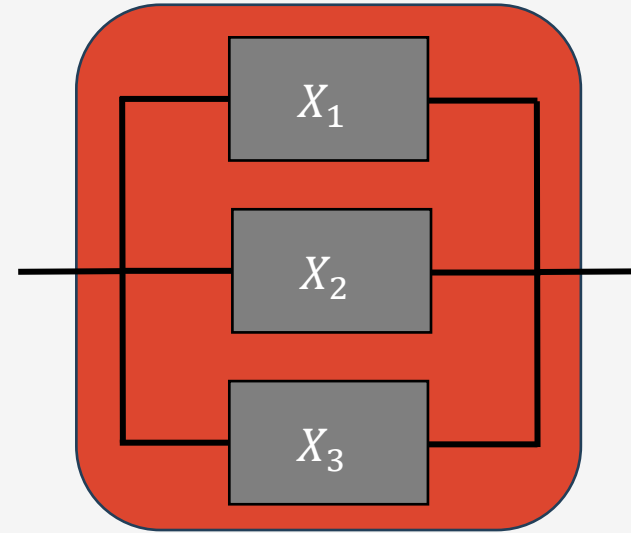
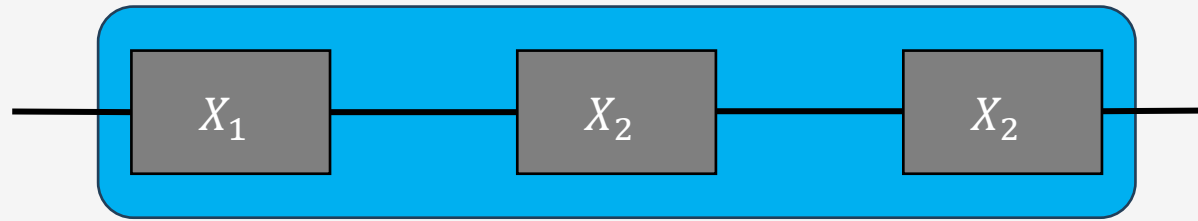
$$\mathbf{1}_{\{L>t\}} = \begin{cases} 1, & \text{if the } \text{system} \text{ is alive} \\ 0, & \text{is dead} \end{cases}$$

$$\mathbf{1}_{\{L_i>t\}} = \begin{cases} 1, & \text{if } \text{component } i \text{ is alive} \\ 0, & \text{is dead} \end{cases}$$

- Then using structure function: $\mathbf{1}_{\{L>t\}} = \varphi(\mathbf{1}_{\{L_1>t\}}, \mathbf{1}_{\{L_2>t\}}, \dots, \mathbf{1}_{\{L_n>t\}})$
- And we obtain the *system Lifetime*: $P\{L > t\} = E[\mathbf{1}_{\{L>t\}}] = E[\varphi(\mathbf{1}_{\{L_1>t\}}, \mathbf{1}_{\{L_2>t\}}, \dots, \mathbf{1}_{\{L_n>t\}})]$

Remark: If the components are connected **in series or in parallel**, then we can use the properties of Expectation and Independent RVs to compute $E[\mathbf{1}_{\{L>t\}}]$ in a computationally efficient way.

Gallery



Gallery #2 : 2 out of 3 system diagram

